

Practical Session 1

Topic: Data, Research Design and Descriptive Statistics

1. Getting Familiar with the Dataset

Context: Annual average inflation rate (%) in the Republic of Uzbekistan for 2021-2025 (or the most recent five-year period):

Year	Inflation rate (%)
2021	10.0
2022	12.3
2023	8.8
2024	9.8
2025	7.1

Sorted sequence (ascending): 7.1, 8.8, 9.8, 10.0, 12.3 (n = 5)

2. Measures of Central Tendency

A. Mean ($\bar{x}$)

Add all values and divide the sum by the number of observations.

$$\bar{x} = \Sigma x_i / n = (7.1 + 8.8 + 9.8 + 10.0 + 12.3) / 5 = 48.0 / 5 = 9.6\%$$

- Economic interpretation: Over the last five years, the average annual inflation rate was 9.6%.

B. Median (Me)

The value located in the middle when the observations are arranged in ascending order.

- Sequence: 7.1, 8.8, 9.8, 10.0, 12.3
- Middle value: 9.8%
- Economic interpretation: 50% of the years had inflation below 9.8%, while the remaining 50% had inflation above 9.8%.

C. Mode (Mo)

The value that occurs most frequently.

- Each value appears only once, so this dataset has no mode.

R code:

```
inflation <- c(10.0, 12.3, 8.8, 9.8, 7.1)

mean(inflation)    # Mean -> 9.6

median(inflation)  # Median -> 9.8
```

3. Measures of Dispersion

A. Sample Variance (s^2)

Measures how far the observations are spread from the mean.

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Calculation table:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
7.1	-2.5	6.25
8.8	-0.8	0.64
9.8	+0.2	0.04
10.0	+0.4	0.16
12.3	+2.7	7.29
Total (Σ)	0	14.38

$$s^2 = 14.38 / (5 - 1) = 14.38 / 4 = 3.595$$

B. Standard Deviation (s)

Take the square root of the variance so that the result remains in the original unit.

$$s = \sqrt{s^2} = \sqrt{3.595} \approx 1.896\%$$

- Economic interpretation: Inflation values deviated from the mean (9.6%) by approximately 1.9 percentage points on average.

R code:

```
var(inflation) # Variance -> 3.595
sd(inflation)  # Standard deviation -> 1.896048
```

4. Five-Number Summary and Boxplot

The five-number summary describes how the data are distributed:

1. Minimum (Min): 7.1
2. First quartile (Q1): 25% point of the data. Position: $(n + 1) \times 0.25 = 1.5 \rightarrow (7.1 + 8.8) / 2 = 7.95$
3. Median (Q2): 9.8
4. Third quartile (Q3): 75% point of the data. Position: $(n + 1) \times 0.75 = 4.5 \rightarrow (10.0 + 12.3) / 2 = 11.15$
5. Maximum (Max): 12.3

Interquartile range (IQR): $Q3 - Q1 = 11.15 - 7.95 = 3.2\%$

R code:

```
summary(inflation)

# Min. 1st Qu. Median Mean 3rd Qu. Max.
# 7.10  8.80    9.80  9.60 10.00  12.30

IQR(inflation)

boxplot(inflation, main="Inflation Rate Boxplot", col="lightblue")
```

R may use a different quartile calculation algorithm.

5. Identifying Outliers

An outlier is an observation that differs sharply from the other values in a dataset. The interquartile range (IQR) rule can be used to identify outliers.

Step-by-step calculation:

1. Calculate IQR: $IQR = Q3 - Q1$
2. Lower bound: $Q1 - 1.5 \times IQR$
3. Upper bound: $Q3 + 1.5 \times IQR$

Rule: Any value below the lower bound or above the upper bound is classified as an outlier.

Example using the inflation data 7.1, 8.8, 9.8, 10.0, 12.3:

$Q1 = 7.95$, $Q3 = 11.15$

$IQR = 11.15 - 7.95 = 3.2$

Lower bound: $7.95 - 1.5 \times 3.2 = 3.15$

Upper bound: $11.15 + 1.5 \times 3.2 = 15.95$

Conclusion: All values (7.1 to 12.3) lie within [3.15, 15.95], so this dataset contains no outliers.

R code:

```
# Test dataset
x <- c(2.0, 7.4, 5.7, 6.0, 6.4, 12.5) # 12.5 is an intentionally added outlier
hist(x, main="1. Histogram", xlab="Values", col="skyblue", border="white")
boxplot(x, main="2. Boxplot", col="gold", horizontal=TRUE)
plot(density(x), main="3. Density Plot", col="red", lwd=2)
dotchart(x, main="4. Dotchart", color="darkgreen", pch=19)
stem(x)
plot(ecdf(x), main="6. Empirical CDF (ECDF)", col="blue", lwd=2)
```

Practical Assignment for Students

Task: Using the five-year economic indicator assigned to you, complete the following:

1. Calculate the mean ($\bar{x}$), median (Me), and mode (Mo).
2. Calculate the sample variance (s^2) and standard deviation (s) manually using the formulas.
3. Find the five-number summary (Min, Q1, Median, Q3, Max) and the interquartile range (IQR).
4. Identify outliers: use the IQR rule to calculate the lower and upper bounds and determine whether the dataset contains any outliers.
5. Work in R: verify your calculations using `mean()`, `median()`, `sd()`, `var()`, `summary()`, and `IQR()`; then draw a histogram (`hist`), boxplot (`boxplot`), and density plot (`density`) for your dataset.

25 Individual Variants (Source: stat.uz / data.egov.uz)

Variant / Student	Indicator and unit	2020	2021	2022	2023	2024
Variant 1	GDP growth rate (%)	2.0	7.4	5.7	6.0	6.4
Variant 2	Annual inflation rate (%)	11.1	10.0	12.3	8.8	9.8
Variant 3	Industrial production growth (%)	0.7	9.5	5.2	6.0	6.8
Variant 4	Agricultural output growth (%)	2.9	4.0	3.6	4.1	4.2
Variant 5	Market services volume growth (%)	2.3	19.2	15.9	13.7	12.8
Variant 6	Construction work volume growth (%)	9.5	6.8	6.6	6.4	8.1
Variant 7	Retail trade turnover growth (%)	3.2	12.1	12.3	10.2	9.2
Variant 8	Foreign trade turnover (billion USD)	36.3	42.1	50.0	62.6	65.8
Variant 9	Export volume (billion USD)	15.1	16.6	19.3	24.4	26.1
Variant 10	Import volume (billion USD)	21.2	25.5	30.7	38.2	39.7
Variant 11	Fixed capital investment growth (%)	-4.4	2.9	0.9	22.1	15.3
Variant 12	GDP per capita (million UZS)	16.9	21.2	25.8	30.3	35.1
Variant 13	Unemployment rate (%)	10.5	9.6	8.9	6.8	5.7
Variant 14	Average monthly nominal wage (million UZS)	2.6	3.2	3.9	4.6	5.2
Variant 15	Real growth of total household income (%)	3.1	11.6	9.7	7.0	6.8
Variant 16	Interregional freight transport volume growth (%)	0.8	11.3	5.1	6.2	7.0
Variant 17	Passenger transport volume growth (%)	-5.1	6.2	5.5	5.7	6.1
Variant 18	Share of small business in GDP (%)	53.9	54.9	51.8	51.2	52.4
Variant 19	State budget deficit-to-GDP ratio (%)	4.5	5.6	3.9	5.5	4.0
Variant 20	Foreign direct investment (billion USD)	2.3	2.5	3.3	4.2	4.8
Variant 21	Food inflation (%)	15.3	13.0	15.6	9.7	2.4
Variant 22	Non-food inflation (%)	8.8	7.8	10.7	7.7	8.1
Variant 23	Services inflation (%)	7.1	8.2	8.4	8.7	27.2
Variant 24	Commercial bank deposits (trillion UZS)	114	156	217	242	285
Variant 25	Population growth rate (%)	2.0	2.1	2.1	2.2	2.1